

Written Report – 6.419x Module 4

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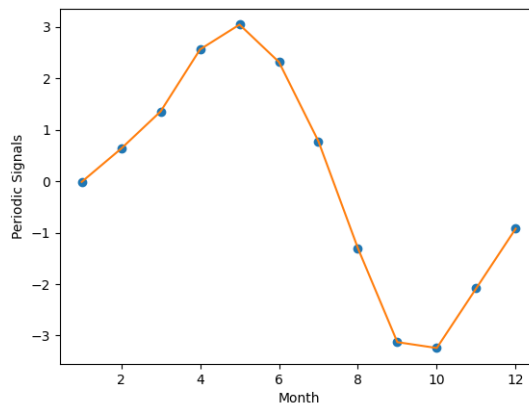
April 2023

2. The Mauna Loa CO_2 Concentration

1. (3 points) Plot the periodic signal . (Your plot should have 1 data point for each month, so 12 in total). Clearly state the definition the , and make sure your plot is clearly labeled.

Solution: We chose the quadratic model for our deterministic trend function, $F_n(t)$. We define our periodic signal of each month, as the average residual $C_i - F_n(t)$ for each month. We collected every residual for each month by removing deterministic trend and then calculating its average. We have the following plot:

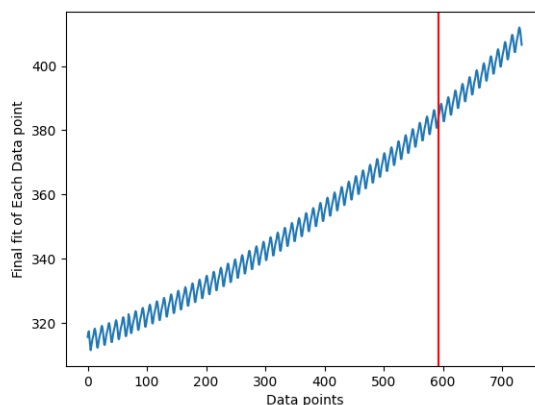
Figure 1: Periodic Signal of Each Month



2. (2 points) Plot the final fit $F_n(t_i) + P_i$. Your plot should clearly show the final model on top of the entire time series, while indicating the split between the training and testing data.

Solution: We have the following plot of the final fit $F_n(t_i) + P_i$ over the entire time series. The red vertical line separates the training data (left of the line) and the test data (right of the line):

Figure 2: Plot of the Final Fit $F_n(t_i) + P_i$



3. (4 points) Report the root mean squared prediction error RMSE and the mean absolute percentage error MAPE with respect to the test set for this final model. Is this an improvement over the previous model $F_n(t_i)$ without the periodic signal? (Maximum 200 words.)

Solution: For our final model $F_n(t_i) + P_i$, we have RMSE ≈ 1.143 and MAPE $\approx 0.2075\%$. For our previous quadratic model $F_n(t_i)$, we have RMSE ≈ 2.501 and MAPE $\approx 0.5320\%$. Since our final model's RMSE and MAPE are both lower than the previous model's counterparts, we determine that our final model $F_n(t_i) + P_i$ is an improvement over the previous model.

4. (3 points) What is the ratio of the range of values of F to the amplitude of P_i and the ratio of the amplitude of P_i to the range of the residual R_i (from removing both the trend and the periodic signal)? Is this decomposition of the variation of the CO_2 concentration meaningful?

Solution: We have the ratio of the range of F to amplitude of $P_i = 95.47 : 3.1429$ and the ratio of the amplitude of P_i to the range of the residual $R_i = 3.1429 : 4.716$. Essentially, this decomposition indicates that while the calculations of F are the main components of CO_2 predictions, P_i and R_i are each smaller components, which when together with F , add up to a more accurate prediction of CO_2 levels. Notice that the range of F is much greater than the amplitude of P_i , while the amplitude of P_i is approximately equal to that of the range of R_i . This is because P_i , which are seasonal variations, and

R_i , which accounts for other residual components, are both smaller influences of CO_2 level predictions – the bulk of which are modelled by our regression model, F . However, P_i and R_i , although small in values, are important parts of our model to accurately predict future CO_2 levels.

3. Autocovariance Functions

1. (4 points) Consider the $MA(1)$ model,

$$X_t = W_t + \theta \cdot W_{t-1}, \quad (1)$$

where $\{W_t\} \sim W \sim \mathcal{N}(0, \sigma^2)$. Find the autocovariance function of $\{X_t\}$.

Solution: By definition, we have that the autocovariance function $\gamma_X(s, t) = \text{cov}(X_s, X_t)$. Since we have $\mu = 0$, we have to calculate $\gamma_X(t, t)$ and $\gamma_X(t, t-1)$, as autocovariance values of $\gamma_X(t, t-2), \gamma_X(t, t-3), \gamma_X(t, t-4), \dots = 0$. We have the following with $X_t = W_t + \theta \cdot W_{t-1}$:

$$\begin{aligned} \gamma_X(t, t) &= \text{cov}(X_t, X_t) = \mathbb{E}[(X_t - \mu)^2] = \mathbb{E}[X_t^2 - 2\mu X_t + \mu^2] \\ &= \mathbb{E}[W_t^2 + 2\theta \cdot W_t W_{t-1} + \theta^2 W_{t-1}^2 - 2\mu(W_t + \theta \cdot W_{t-1}) + \mu^2] \\ &= \sigma^2 + \theta^2 \sigma^2 \end{aligned} \quad (2)$$

since $\mu = 0$ and therefore $\mathbb{E}[-2\mu(W_t + \theta \cdot W_{t-1})] = 0$; and for all t , $\mathbb{E}[W_t^2] = \sigma^2$, and $\mathbb{E}[W_t] = 0 \Rightarrow \mathbb{E}[2\theta \cdot W_t W_{t-1}] = 0$. Next, we find $\gamma_X(t, t-1)$:

$$\begin{aligned} \gamma_X(t, t-1) &= \text{cov}(X_t, X_{t-1}) = \mathbb{E}[(W_t + \theta \cdot W_{t-1}) \cdot (W_{t-1} + \theta \cdot W_{t-2})] \\ &= \mathbb{E}[W_t W_{t-1} + \theta \cdot W_t W_{t-2} + \theta \cdot W_{t-1}^2 + \theta^2 W_{t-1} W_{t-2}] \\ &= \theta \sigma^2 \end{aligned} \quad (3)$$

We have $\gamma_X(t, t) + \gamma_X(t, t-1) = \sigma^2 + \theta^2 \sigma^2 + \theta \sigma^2 = (1 + \theta + \theta^2) \sigma^2$.

2. (4 points) Consider the $AR(1)$ model,

$$X_t = \phi \cdot X_{t-1} + W_{t-1}, \quad (4)$$

where $\{W_t\} \sim W \sim \mathcal{N}(0, \sigma^2)$. Suppose $\|\phi\| < 1$. Find the autocovariance function of $\{X_t\}$.

Solution: We assume X_t is stationary if $\|\phi\| < 1$. By definition, we have that the autocovariance function $\gamma_X(s, t) = \text{cov}(X_s, X_t)$. Then, we have with mean μ :

$$\begin{aligned} \gamma_X(t, t-1) &= \text{cov}(X_t, X_{t-1}) = \text{cov}(\phi \cdot X_{t-1} + W_{t-1}, X_{t-1}) \\ &= \phi \cdot \text{cov}(X_{t-1}, X_{t-1}) + \text{cov}(W_{t-1}, X_{t-1}) \\ &= \phi \gamma_X(t-1, t-1) = \phi \cdot \text{Var}(X_{t-1}) \end{aligned} \quad (5)$$

In our solution, we used the property for real-valued random variables X, Y, W, V with real-valued constants a, b, c, d , $cov(aX+bY, cW+dV) = ac \cdot cov(X, W) + ad \cdot cov(X, V) + bc \cdot cov(Y, W) + bd \cdot cov(Y, V)$ and that for white noise, $cov(W_{t-1}, X_{t-1}) = 0$.

5. Converting to Inflation Rates

*Repeat the model fitting and evaluation procedure from the previous page for the monthly inflation rate computed from **CPI**.*

Your response should include:

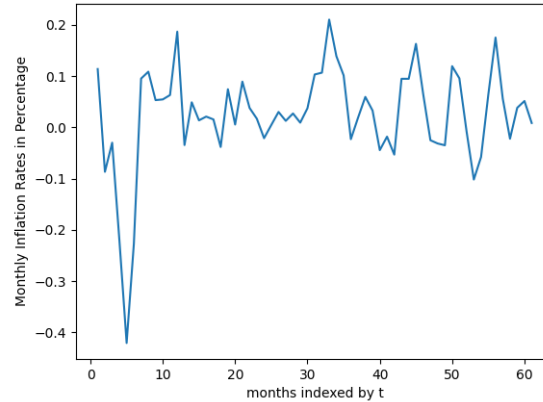
- *(1 point) Description of how you compute the monthly inflation rate from **CPI** and a plot of the monthly inflation rate. (You may choose to work with log of the CPI.)*
- *(2 points) Description of how the data has been detrended and a plot of the detrended data.*
- *(3 points) Statement of and justification for the chosen $AR(p)$ model. Include plots and reasoning.*
- *(3 points) Description of the final model; computation and plots of the 1 month-ahead forecasts for the validation data. In your plot, overlay predictions on top of the data.*

Solution: In our CPI data, the first and last entry contain empty values and will be omitted for our calculations of monthly inflation rates. First, we will take the logarithm transform over the inflation data. Then, we will compute the monthly inflation rates via the following equation:

$$IR_t = \frac{CPI_t - CPI_{t-1}}{CPI_{t-1}} \quad (6)$$

where t indexes the months. We will multiply 100 to this value, to calculate the monthly inflation rate in percentages. We have the following plot of monthly inflation rates, over t in the next page:

Figure 3: Plot of Monthly Inflation Rates



In our plot, we observe no trend of our inflation rate data. Indeed, this is likely because our calculation for monthly inflation rates removes the linear trend present in our CPI data. Observe that in equation (6), we subtract current CPI value by its previous counterpart – the result of which is divided by the same counterpart. In essence, by computation of our monthly inflation rates, we remove the trend of our CPI values.

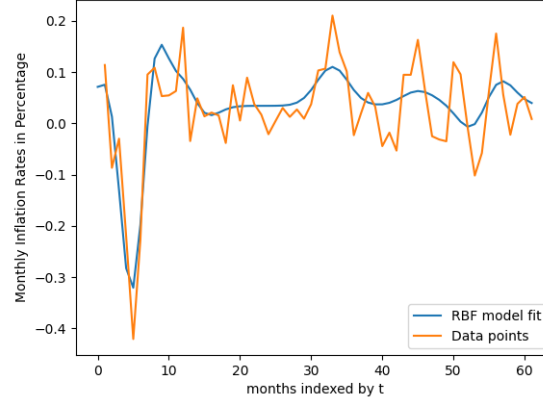
However, observe that our data contains outliers as well, in the initial parts of the plot. We address this by fitting an Radial Basis Function (RBF) Kernel via the following code:

```
1 from sklearn.svm import SVR
2 svr_rbf = SVR(kernel = "rbf", gamma = 0.1, epsilon = 0.1)
3 svr_rbf.fit(np.array(X_train).reshape(-1, 1), y_train)
4
5 cpi_logIR_pred = svr_rbf.predict(np.array(X_train).reshape(-1, 1))
6 plt.plot(cpi_logIR_pred, label = 'RBF model fit')
7 plt.plot(ir, label = 'Data points')
8 plt.legend(loc="lower right")
9 plt.xlabel("months indexed by t")
10 plt.ylabel("Monthly Inflation Rates in Percentage")
```

Listing 1: RBF kernel fitted on Monthly Inflation Rate Data

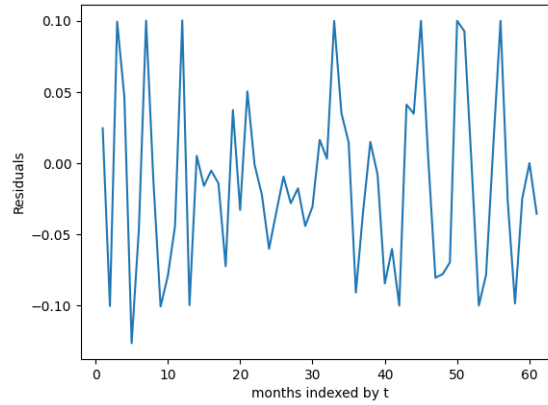
We get the following plot in the next page:

Figure 4: Plot of RBF Kernel Model on Monthly Inflation Rate Data



Removing the trend gives the following plot of the residual $R_t = IR_t - F_t$, where IR_t is the logarithm-transformed inflation rate at month index t and F_t is the fitted RBF kernel value at t :

Figure 5: Data with Trend Removed



Since our data points do not contain seasonal variation, we move on to fitting an auto regression model on the residual $R_t = IR_t - F_t$. In order to determine the lag input of our auto regression model, we first compute the auto-correlation of R_T via the following code.

```

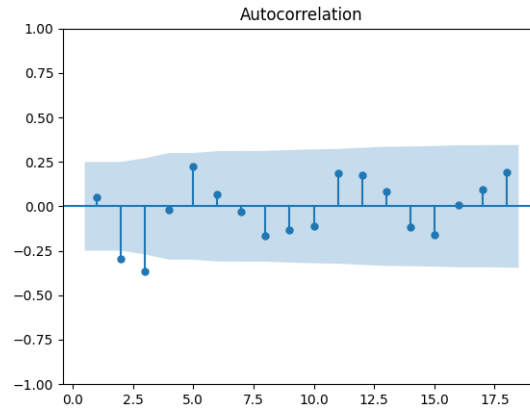
1 import statsmodels.api as sm
2 sm.graphics.tsa.plot_acf((y_train - cpi_logIR_pred), zero = False)

```

Listing 2: Code for Auto-correlation of R_t

We have the following plot in the next page:

Figure 6: Plot of Auto-correlation of R_t



Observe, in the plot, that the maximum lag where auto-correlation value exceeds the observed boundary of statistical significance, is at lag 3. Hence, we choose lag $p = 3$ for our auto-regression model. We have the following code for our auto-regression model:

```

1 from statsmodels.tsa.ar_model import AutoReg
2 # y_train = cpi_train['log IR']
3 res = AutoReg((y_train - cpi_logIR_pred), lags = 3).fit()
4 res.params

```

Listing 3: Code for Auto-regression Model of R_t

which gives the following coefficients:

```

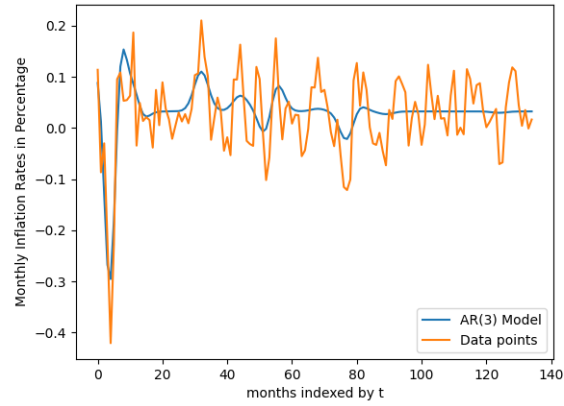
1 const      -0.022923
2 log IR.L1   0.061977
3 log IR.L2  -0.349159
4 log IR.L3  -0.302239
5 dtype: float64

```

Listing 4: Coefficients of Auto-regression Model of R_t

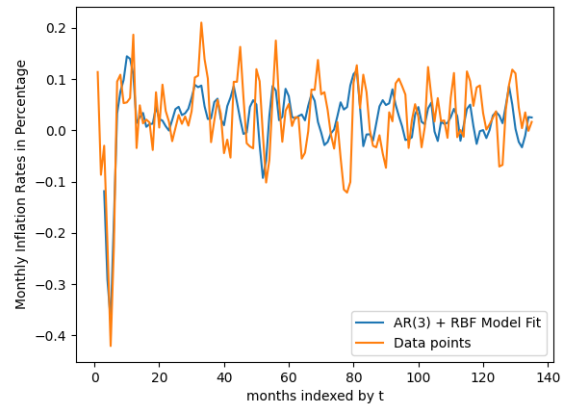
Using this $AR(3)$ model, we get the following prediction on validation data in the next page, computed with the above constant and coefficients in Listing 4:

Figure 7: $AR(3)$ Model Fit over Entire Data



Notice that the trend decays over time. Indeed, this is because our $AR(3)$ model does not account for the trend modelled by our RBF kernel. Simply adding RBF kernel values for corresponding $AR(3)$ values will provide a much more accurate fit, without decay. We have the following plot of 1-month ahead validation predictions made by using all the data before that month :

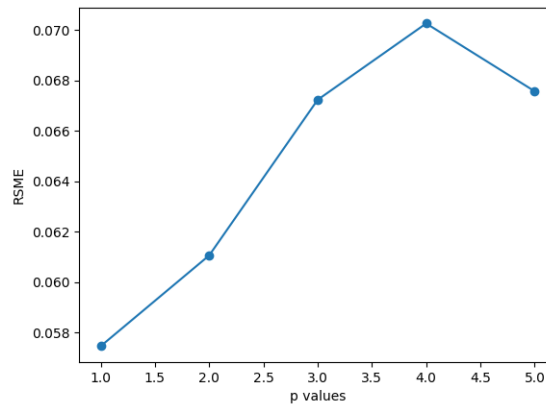
Figure 8: $AR(3) + RBF$ Model Fit over Entire Data



2. (3 points) Which $AR(p)$ model gives the best predictions? Include a plot of the RMSE against different lags for the model.

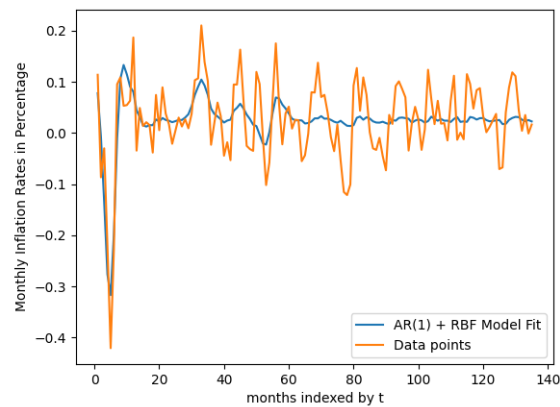
Solution: For this problem, we will examine p values of 1, 2, 3, 4, 5. We have the following plot of RMSE's over these values of p :

Figure 9: RMSE for Different p Values



By our plot, $p = 1$ gives the lowest RMSE, which could indicate that $AR(1)$ gives the most optimal predictions. However, observe the model plot for $p = 1$:

Figure 10: $AR(1)$ + RBF Model Fit over Entire Data



Notice that the $AR(1)$ model, although it captures trends in the data, the model does not fit well to the data. Below, are model fits for $p = 2, 4, 5$:

Figure 11: $AR(2) + RBF$ Model Fit over Entire Data

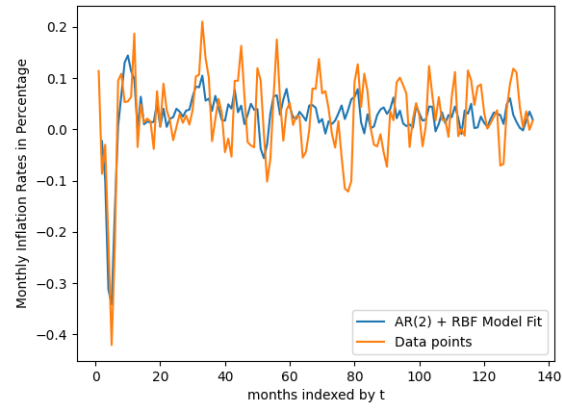


Figure 12: $AR(4) + RBF$ Model Fit over Entire Data

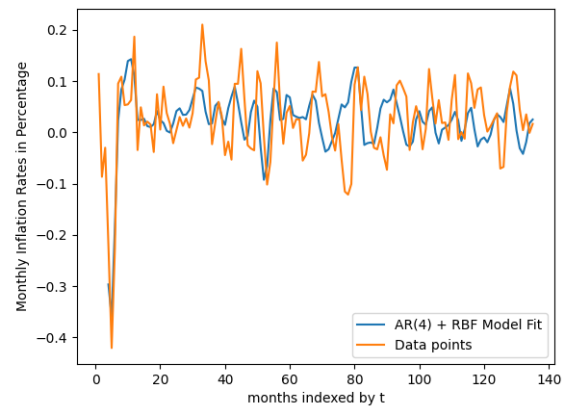
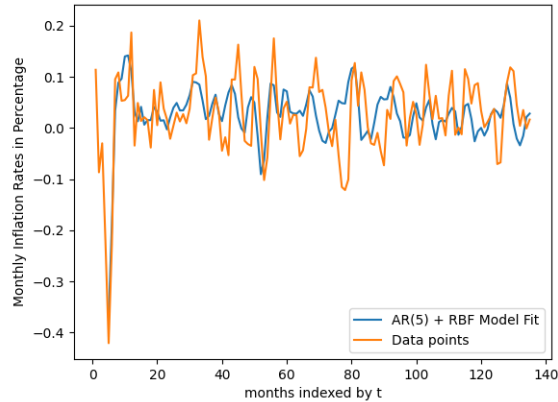


Figure 13: $AR(5) + RBF$ Model Fit over Entire Data



Observe that the $AR(4)$ and $AR(5)$ models fit just as well to the data, as the $AR(3)$ model, while the $AR(2)$ model is comparatively worse. Numerically $AR(3)$, $AR(4)$ and $AR(5)$ have approximate RMSE values of 0.06724, 0.07027, and 0.06758 respectively. Since among these 3 the $AR(3)$ model has the lowest RMSE, we conclude that the $AR(3)$ model gives the best predictions.

(3 points) Overlay your estimates of monthly inflation rates and plot them on the same graph to compare. (There should be 3 lines, one for each datasets, plus the prediction, over time from September 2013 onward.)

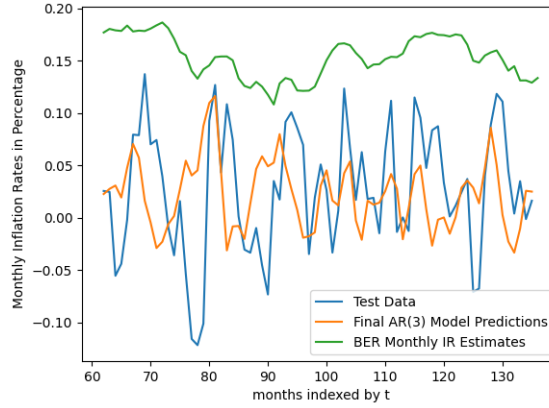
Solution: Here, our monthly BER inflation estimates were calculated via the following code:

```
1 # BER_Month['BER Avg'] represents the average BER values for each
   month
2 # BER_Month collects the average BER for each month into a new data
   frame
3 BER_Month['BER IR'] = ((BER_Month['BER Avg'] / 100 + 1) ** (1 / 12)
   - 1) * 100
```

Listing 5: Monthly BER Inflation Rate Estimates

We have the following plot in the next page:

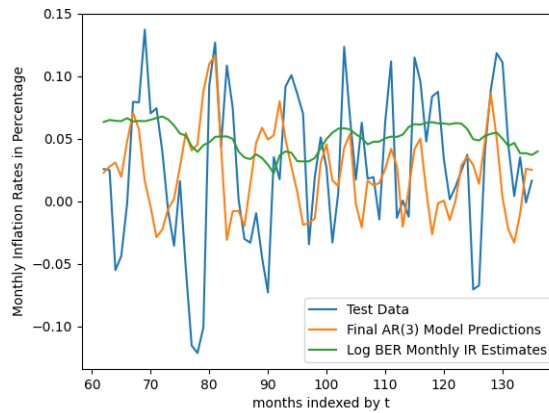
Figure 14: BER, Test and $AR(3)$ Comparison



Notice that the BER monthly inflation rate estimations are not aligned with the test and prediction data. This is because when we computed our predictions with our test data, we used a logarithm transformed before the computations. Observe that BER plot follows the trend in the test data, and the model predictions.

If we take the logarithm transform of the BER values before removing their annual trends to get the monthly BER inflation rates, we get the following plot:

Figure 15: Log BER, Test and $AR(3)$ Comparison



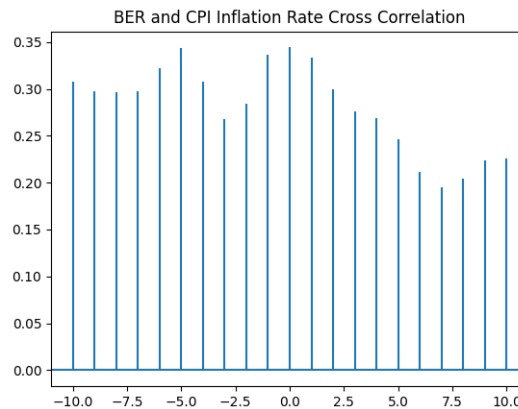
Notice that our logarithm transformed BER data captures trends and variations, as much as its counterpart that was not logarithm transformed. As expected, our test data, $AR(3) + RBF$ model predictions, and the logarithm transformed BER monthly inflation rates align much better. If we remove the logarithm transforms for all of the data used, we will get a similar plot, but with slightly different values in the y -axis.

6. External Regressors and Model Improvements (Written Report)

1. (4 points) Plot the cross correlation function between the CPI and BER inflation rate, by which find r , i.e., the lag between two inflation rates. (As only one external regressor term is involved in the model, we only consider the peak in the CCF plot.)

Solution: For this problem, when calculating monthly inflation rates from CPI, we will use the original data; for fair comparison, we will not use the logarithm transformed data from the previous problem. We have the following plot:

Figure 16: BER and CPI Inflation Rate Cross Correlation

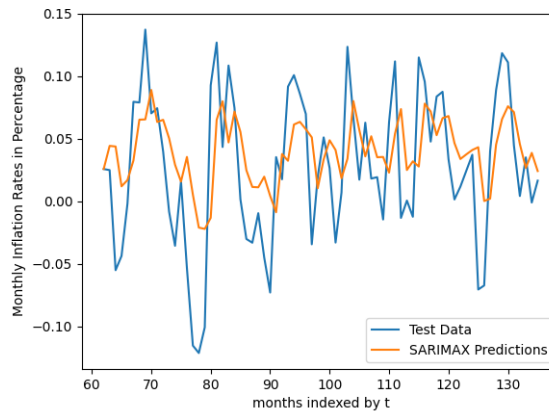


From our plot, we determine that $r = 0$ and $r = -5$ to be the lags between the 2 inflation rates, since the highest peaks in the graph corresponds to 0 and -5 in the x -axis.

2. (3 points) Fit a new AR model to the CPI inflation rate with these external regressors and the most appropriate lag. Report the coefficients, and plot the 1 month-ahead forecasts for the validation data. In your plot, overlay predictions on top of the data.

Solution: For this problem, we will take $r = 0$. We will also use the monthly inflation rate calculated with logarithm transformed values of CPI; we will keep our original BER values. Instead of the auto-regression model used before, we will use Seasonal Auto-Regressive Integrated Moving Average with Exogenous regressors model. We have the following plot of the 1 month-ahead forecasts for the validation data in the next page:

Figure 17: SARIMAX Model, Logarithm Transformed CPI, Original BER



3. (3 points) Report the mean squared prediction error for 1 month ahead forecasts.

Solution: For our SARIMAX model, our mean squared error, without taking its root, was valued at 0.0027789614162847414 by the following code:

```
1 # y_test = cpi_test['IR']
2 # sar_test_pred represents the validation predictions of SARIMAX
3 SARIMAX_mse = mean_squared_error(y_test.values[:-2], sar_test_pred
4                                  [:-2])
5 print(SARIMAX_mse)
```

Listing 6: MSE of SARIMAX Model

The error's rooted version, the RMSE, is 0.052715855454357766 given by the following:

```
1 # y_test = cpi_test['IR']
2 # sar_test_pred represents the validation predictions of SARIMAX
3 SARIMAX_mse = mean_squared_error(y_test.values[:-2], sar_test_pred
4                                  [:-2], squared = False)
5 print(SARIMAX_mse)
```

Listing 7: RMSE of SARIMAX Model

(5 points) (5 points) What other steps can you take to improve your model from part III? What is the smallest prediction error you can obtain? Describe the model that performs best. You might consider including MA terms, adding a seasonal AR term, or adding multiple daily values (or values from different months) of BER data as external regressors.

Solution: In this section, we will transform the BER data via the natural logarithm, make prediction plots with the new SARIMAX model with the logarithm transformed BER data, and collect its RMSE value for comparison. We will also take the same approach, for SARIMAX model without any logarithm transformations – neither on the CPI data, nor on the BER data. Ultimately, we will draw conclusions about the necessity of scaling for the data provided to us.

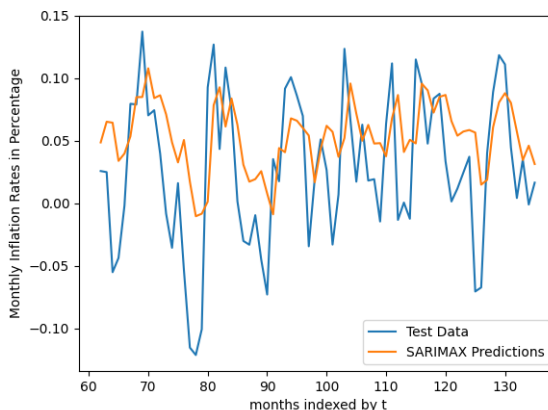
First, we edit our monthly BER inflation rate calculations via the following code:

```
1 # BER_Month['BER Avg'] represents the average BER values for each
  month
2 # BER_Month collects the average BER for each month into a new data
  frame
3 BER_Month['BER IR'] = ((np.log(BER_Month['BER Avg']) / 100 + 1) **
  (1 / 12) - 1) * 100
```

Listing 8: Monthly Logarithm BER Inflation Rate Estimates

With the monthly BER inflation rates calculated from logarithm transformed BER values, we have the following plot and RMSE value 0.057526965303063:

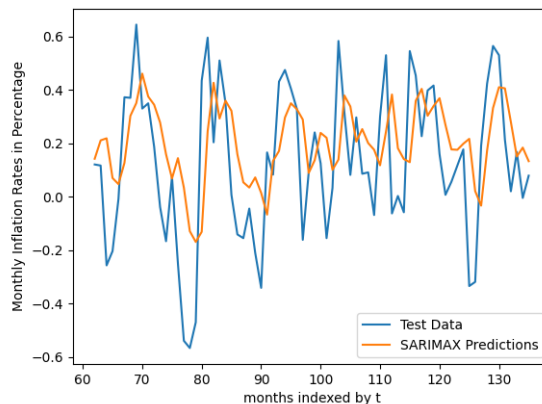
Figure 18: SARIMAX Model, Log Transformation on All Data



The SARIMAX predictions with logarithm transformed monthly BER inflation rate estimates have RMSE just slightly less than what we found in the

previous section. Now, we will model SARIMAX without any logarithm transformation. We have the following plot and RSME value 0.25878559307356885:

Figure 19: SARIMAX Model, No Log Transformation



Notice that with our model, without scaled data, has a much higher comparative RMSE score. This indicates that our data would benefit from scaling, as we expected; however, we also observed that our previous model, where CPI and BER data underwent logarithm transformation, it performed worse than the model where BER was not transformed. This is likely because unlike CPI data, BER does not benefit very much from a logarithm transformation, which indicates that original BER data points are not as are not much different from their transformed counterparts; the original points are not unnecessarily very far apart from each other. Indeed, observe the following plots in the next 2 pages:

Figure 20: Original BER Data Points

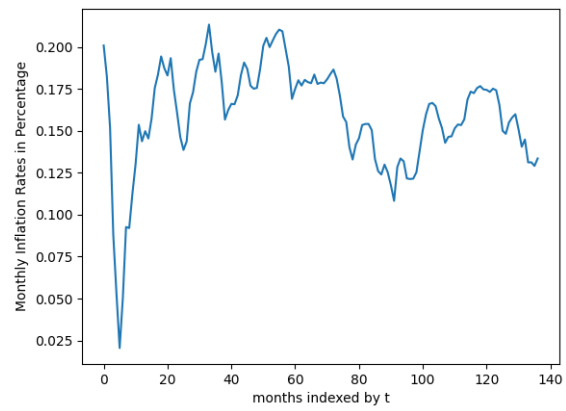


Figure 21: BER Data Points with Logarithm Transformation

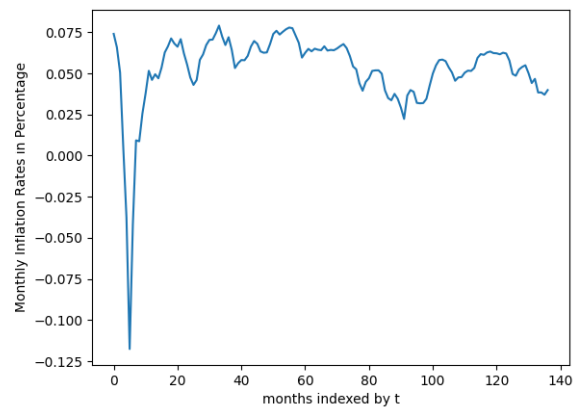


Figure 22: Original CPI Data Points

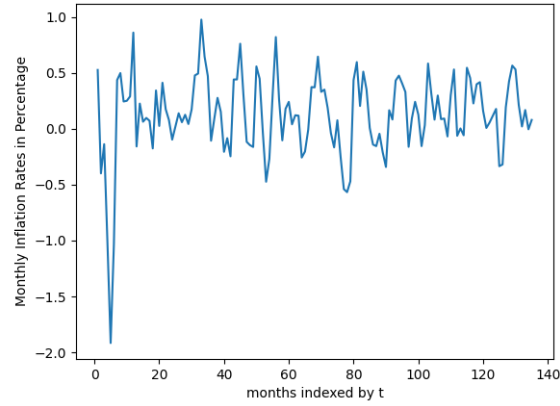
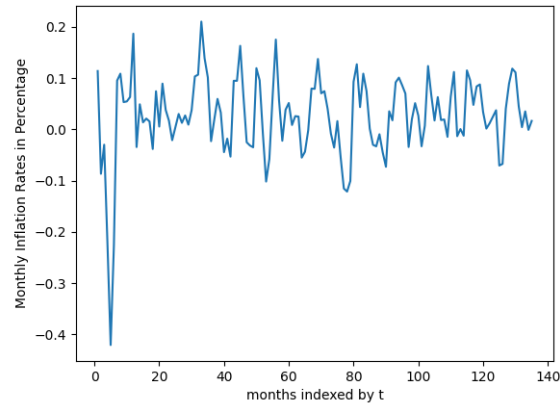


Figure 23: CPI Data Points with Logarithm Transformation



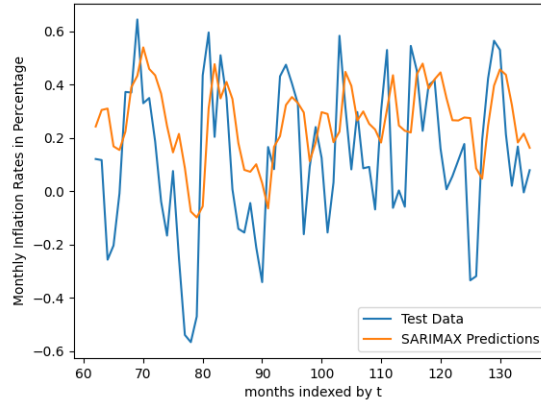
When logarithm transformed, the shape of BER data changes very noticeable; points are very squished together, their distance parameters mutated. This can explain why the SARIMAX model with transformations on BER data performed worse. Through the transformation, the model interprets BER data differently, and for this data set, a transformation gives worse results.

Observe CPI data points; they maintain their general shapes and trends. Applying a standardization method, such as logarithm transformation, would be

very beneficial in this case, because the distance parameters between each point become scaled, with any unnecessary exaggerations in distance removed via the logarithm transformation. Our model would interpret our data better from logarithm transformation. This is also reflected in our RMSE scores calculated earlier. The model with the lowest RMSE was our SARIMAX model with transformed CPI data, and original BER data; the worst model was SARIMAX model with original CPI and BER data points. This corroborates our conclusion that the transformed CPI data performs much better.

On a final note, we create a SARIMAX model with original CPI data and logarithm transformed BER data. We have the following plot and RMSE value 0.2804060064915618:

Figure 24: SARIMAX Model, Original CPI, Logarithm Transformed BER



As expected, we have the worst RMSE score; what we did here, remove the benefits of logarithm transformation on CPI data, and apply the transformation on BER data, which changed its structure. This further implies that while logarithm transformation on CPI data is extremely useful, the exact opposite applies to BER data.

In conclusion, the best representative of model predictions is given by the SARIMAX model with endogenous logarithm transformed CPI data, and exogenous original BER data, with $RMSE = 0.052715855454357766$. RMSE scores computed from $AR(p) + RBF$ models from earlier perform just slightly worse, which likely means that the exogenous BER data is beneficiary in getting better predictions on monthly inflation rate data.